

PIA 403 Course Outline

PIA 403 – Financial Appraisal of Public Projects

What the money does: cash flow, cover, debt and the funding gap.

School	04 – Public Investment
Hours	30 (indicative band 27-33)
Version	1.0 (2026-09-05)
Model	AICTEM PIAS, D:/EDIP/models/PIASModelling
Reference country	Utopia (UTP), UTD 2024 prices
Prerequisites	PIA 401 for the cycle, the conventions and the governed parameter set. PIA 402 is useful and not required. Arithmetic at the level of a compounding factor; no calculus.
Audience	Project finance officers, treasury and debt management staff, utility and agency finance departments, and appraisal economists who have to read a financial model somebody else built
Pass mark	60 %

What this course is

The financial appraisal asks one question: what does the money do? It works in market prices, includes taxes, counts the loan and the repayment, and answers whether the project pays for itself and whether it can service its debt in every single year.

It is routinely confused with the economic appraisal, which asks a different question in different prices, and the confusion runs both ways. An analyst who reports a negative FNPV as a finding has misunderstood public investment: roads do not sell tickets, and 11 of the eleven sample projects have a negative FNPV. An analyst who omits the financial appraisal because the project is public has misunderstood something worse, because the debt is real, the tariff is real, and the year in which the cover ratio falls below one is the year the project defaults.

The course covers discounting properly, including the internal rate of return and the cases in which PIAS refuses to report one; the financial cash flow and the tax that applies to operating surplus rather than to revenue; the funding gap; the debt schedule with its grace, its fees and its two conventions for grace interest; the grant element that tests whether a loan called concessional is; and the debt service cover ratio, whose minimum is the number that matters.

Its central discipline is that an average conceals a default. A project with an average cover of 1.8 and a single year at 0.7 defaults in that year, and the average is the number most often reported.

What a graduate can do

Build a financial cash flow from a cost profile and a revenue forecast, applying tax to operating surplus rather than to revenue; compute FNPV, FIRR, payback and discounted payback; state the conditions under which an IRR exists and recognise the three answers PIAS gives; compute a funding gap and a funding gap ratio and say what a ratio of 1.0 means; build a debt schedule with grace, commitment and front-end fees; distinguish the two grace interest conventions and quantify the difference; compute a grant element and test a claim of concessionality against the 25 per cent threshold; compute a debt service cover ratio year by year and identify the minimum; and explain why a negative FNPV is normal for public investment while a minimum DSCR below one is not.

What this course does not claim

This course does not compute an economic case. It works in market prices throughout: taxes are included, transfers are counted, and the discount rate is the financial one, 12.0%, not the social one. PIA 404 does the economic case and PIA 406 supplies the conversion factors that separate them.

It does not teach public-private partnerships beyond the point where they touch the financial appraisal. PIAS implements a PPP contingent-liability path and this course names it; the risk allocation, the availability payment structure and the fiscal commitment accounting are not covered here.

It does not supply a national borrowing strategy, a debt sustainability analysis, or an exchange-rate forecast. The debt schedule it builds is a project schedule; the aggregate debt path belongs to FINEX, and when FINEX has not run the portfolio says the macro-fiscal consistency check is NOT RUN.

It does not claim the register's revenue forecasts are good. Every one of the 144 revenue rows rests on **basis = assumption**, and the course treats that as the standing weakness of a financial appraisal rather than as a property of this dataset.

Learning outcomes

1. Distinguish the financial from the economic appraisal, and state what each includes that the other excludes.
2. Build a financial cash flow, applying tax to operating surplus rather than to revenue.
3. Compute a net present value, an annuity factor and an equivalent annual cost.
4. State the conditions under which an internal rate of return exists, and recognise the three answers PIAS gives.
5. Compute payback and discounted payback, and say why they are not decision rules.
6. Compute a funding gap and a funding gap ratio, and interpret a ratio of 1.0.
7. Build a debt schedule with disbursement, grace, interest, principal and fees.
8. Distinguish the two grace interest conventions and quantify the difference between them.
9. Compute a grant element and test a claim of concessionality against the 25 per cent threshold.
10. Compute a debt service cover ratio year by year, and identify the minimum and the year it falls in.
11. Explain why a negative FNPV is normal for public investment and a minimum DSCR below one is not.
12. State what a financial appraisal does and does not establish about whether a project should be built.

Module structure

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M01	The financial question	1	1.25	–	1	Quiz M01
M02	The financial cash flow	1	3.25	Lab 01	2	Quiz M02
M03	Discounting and the IRR	2	3.50	Lab 02	3-4	Quiz M03
M04	When PIAS refuses to report an IRR	1	1.50	–	5	Quiz M04
M05	Payback, and why it is not a rule	1	1.50	–	6	Quiz M05
M06	Revenue, tariffs and the funding gap	1	3.50	Lab 03	7	Quiz M06
M07	The debt schedule	2	3.50	Lab 04	8-9	Quiz M07
M08	Grace interest, both conventions	1	1.50	–	10	Quiz M08
M09	The grant element and concessionality	1	1.50	–	11	Quiz M09
M10	Debt service cover	1	3.50	Lab 05	12	Quiz M10
M11	Break-even revenue and the tariff question	1	1.50	–	13	Quiz M11
M12	What the financial result establishes	1	1.50	–	14	Quiz M12

Laboratories

Lab	Title	Hours	Deliverable
1	Build a financial cash flow, and check the tax line	2.0	A financial cash flow with the tax error priced

Lab	Title	Hours	Deliverable
2	Discounting, and a cash flow with two roots	2.0	The four operations, and a two-root flow resolved
3	The funding gap, the break-even tariff, and testing the assumption	2.0	A funding gap, a tariff, and the room the appraisal has
4	The debt schedule under both grace conventions, and the grant element	2.0	A schedule under both conventions, and six grant elements
5	Cover, year by year, and the worst year	2.0	A DSCR profile with the worst year named

Assessment

Component	Weight
Knowledge quizzes	15 %
Laboratory exercises	30 %
Applied assignments	25 %
Capstone / final project	30 %

Applied assignments

ID	Title	Weight	Deliverable
A1	The cash flow and the tax line	20	Marked on the tax treatment and the reconciliation to the stored FNPV
A2	The funding gap and the tariff	25	Marked on the tariff derivation and the reported margin
A3	The debt schedule, both ways	30	Marked on the capitalised interest and the concessionality test
A4	Cover, and the worst year	25	Marked on naming the year and on the treatment of NOT AVAILABLE

Final project

A complete financial appraisal of one sample project: the cash flow built from the cost profile and the revenue forecast with tax on operating surplus; FNPV, FIRR with its status, payback and discounted payback; the funding gap and what it implies about who pays; a debt schedule under both grace conventions with the difference quantified; the grant element tested against the concessionality threshold;

the DSCR year by year with the minimum identified and the year named; and a statement of what the financial result does and does not establish about whether the project should be built.

Folder contents

Every course in the Academy carries the same section 5 folder. See `README.md` for what each item holds and `COURSE_BUILD_REPORT.md` for what was measured.